Part 7 Brief · Backend protocol + FHIR R4

mcp-ops-server · 20 Oct 2026 · PR #16 · Status: Codex review fixed (3x P1, 3x P2) + 1 Bugbot; ready to merge

WHAT WAS BUILT

- **One `Backend` protocol** that tools and the eval harness depend on. Today's SQLite store moved behind it unchanged: 179 tests, schema snapshot and eval gate passed before any FHIR code.
- **A FHIR R4 backend** over plain REST (httpx2, no new dependency): PractitionerRole = provider, Schedule + Slot = open slots, Appointment = booking, ServiceRequest = referral. Synthetic seed exported as deterministic transaction Bundles (same IDs, patients are identifiers only); load, reset, export CLI.
- **Parity enforced in CI:** a new `fhir` job runs HAPI (pinned by digest), every tool test on both backends (294 tests), and the eval gate on FHIR. A test asserts the whole eval run (every result, error text, state change and grade) is identical on both.

HOW IT WORKS

Risk	How the FHIR backend handles it
Double booking	Read the sequence first, check everything, then one transaction: Slot busy with If-Match, sequence +1 with If-Match, new Appointment. 409/412 = someone moved: re-check from scratch until the deadline.
Paging	Every search follows every next link (tests use page size 1 and 7); foreign links refused.
Stale search	Cache-Control: no-cache on every request (HAPI caches searches up to 60 s).
Time zones	Slots stored as UTC instants; tools still see clinic-local time (fixed UTC-05:00).
Bad config	Fails closed at start, one line: unreachable, not R4, or no seed loaded.

WHY IT WAS DONE THIS WAY

- **Two guards, each proven alone.** The slot If-Match is the per-slot guard the plan asked for; the sequence keeps APT numbering identical to SQLite and serialises bookings like BEGIN IMMEDIATE, so the patient-overlap check cannot be raced either.
- **Shared error messages** in base.py: an agent sees the same text on either store, so evals compare.

WHAT WENT WRONG OR SURPRISED ME (LOGGED IN FAILURES.MD)

- **A real race, found by breaking my own guard.** With the slot If-Match removed, 20 threads booked one slot 3-5 times: the sequence was read after the slot. Fixed by reading it first. Mutation matrix (15 runs each): either guard alone 0 double bookings, neither 15/15.
- **Codex 3x P1, 3x P2, all reproduced and fixed.** A write that timed out was reported 'cancelled' yet had committed: now it is read back, or reported as 'may or may not' with a safe retry. The deadline bounded each socket wait, not the call (a dribbling server ran 3x over): now enforced on the wait. Also: past slots crowded out upcoming ones (Bugbot too), reset ignored the seed, system clock used host time.

VERIFY IT YOURSELF

```
docker run -d -p 127.0.0.1:8080:8080 hapiproject/hapi:v8.4.0-2
python -m mcp_ops_server.backends.fhir_seed load --url http://127.0.0.1:8080/fhir
MCP_OPS_TEST_BACKENDS=sqlite,fhir python -m pytest -q          # 294 passed
python -m evals.run --backend fhir --fhir-url http://127.0.0.1:8080/fhir  # 31/31, 12/12
```

INTERVIEW TALKING POINT

"The race test passed on the first try, so I removed each guard on purpose. One removal still passed for the wrong reason and exposed a real double-booking window. Now every guard has a test that fails without it."

NEXT (YOUR ACTIONS)

1) Merge PR #16. 2) Add **fhir** to the ruleset's required checks. 3) Say **go** for Part 8.